



Cairo University  
Faculty of Engineering  
Computer Engineering Department

## Machine-Sound Anomaly Detection

Team Number 19:  
Anas Ibrahim Ahmed  
Amr Samy  
AliEldeen Alaa  
Abdelrahman Medhat

*Supervisor: Eng.Ahmed Hashish*

## Abstract

Acoustic anomaly detection is a critical component of predictive maintenance in industrial environments, yet traditional machine learning pipelines frequently suffer from severe data leakage, leading to artificially inflated performance metrics. In this report, we propose a robust, zero-leakage pipeline for Machine-Sound Anomaly Detection. We first address the leakage problem by extracting spatial embeddings from Mel spectrograms using a Convolutional Autoencoder. These embeddings are then clustered using Gaussian Mixture Models (GMM) optimized via the Bayesian Information Criterion (BIC), allowing us to stratify the dataset by independent recording sessions rather than randomized frames. Finally, to handle the inherent class imbalance between normal and abnormal machine states, we deploy a highly-regularized 1D AlexNet-style neural network optimized with a custom Focal Loss function. Evaluated on a strictly segregated, zero-leakage test set, our proposed architecture achieved a highly competitive Macro F1-Score of 0.89, demonstrating exceptional generalization and a strong capacity to isolate anomalous acoustic signatures in real-world conditions.

## Introduction

Predictive maintenance relies heavily on the continuous monitoring of machinery to detect operational faults before they lead to catastrophic failures. Among the various modalities used for monitoring, acoustic anomaly detection offers a non-intrusive and highly effective method for identifying mechanical degradation. However, developing reliable models for machine-sound classification presents significant challenges, primarily due to extreme class imbalances and the risk of data leakage.

Data leakage occurs when highly correlated audio frames from a single continuous recording session are inadvertently distributed across both training and testing datasets. This allows models to achieve near-perfect accuracies simply by memorizing background noise rather than learning true anomalous acoustic features. Early trials in this study utilizing K-Means clustering highlighted these flaws, as rigid, spherical cluster assumptions failed to capture the complex, elongated distributions of audio features, causing the models to fail entirely when tested on truly unseen data.

To bridge this gap, this project aims to develop a robust, zero-leakage acoustic classification system. We introduce a novel pipeline that shifts from traditional statistical feature extraction to spatial feature learning. By converting raw audio into Mel spectrograms and compressing them via a Convolutional Autoencoder, we extract dense 256-dimensional spatial embeddings. We then apply Gaussian Mixture Models (GMM) to accurately identify independent recording sessions, allowing for a strictly stratified, zero-leakage dataset split. Finally, we evaluate multiple architectures—ranging from tuned XGBoost ensembles to advanced Feature Transformers—ultimately deploying a unified 1D AlexNet-style neural network that leverages a custom Focal Loss function to successfully identify anomalies across diverse machine environments.

**Repository Link:** <https://github.com/anas-ibrahem/ml>

## System Overview

The proposed anomaly detection framework operates through a sequential, four-phase pipeline designed to maximize feature clarity and prevent data leakage:

1. **Data Preprocessing:** Raw acoustic signals undergo silence trimming, spectral noise reduction, gain normalization, and high-pass filtering to remove low-frequency mechanical hums.
2. **Feature Engineering:** Cleaned audio is transformed into  $128 \times 128 \times 1$  Log-Mel Spectrograms, mapping frequencies to a human-centric logarithmic scale optimized for convolutional networks.
3. **Zero-Leakage Stratification:** Spatial embeddings are extracted via a Convolutional Autoencoder and clustered using Gaussian Mixture Models (GMM). These clusters represent independent recording sessions and form the boundaries for a strict Train/Validation/Test split, guaranteeing zero data leakage.
4. **Classification:** A highly regularized, 1D AlexNet-style deep neural network processes the structured data. Optimized with a custom Focal Loss function, the model dynamically handles class imbalances to classify the machine state as Normal or Abnormal.

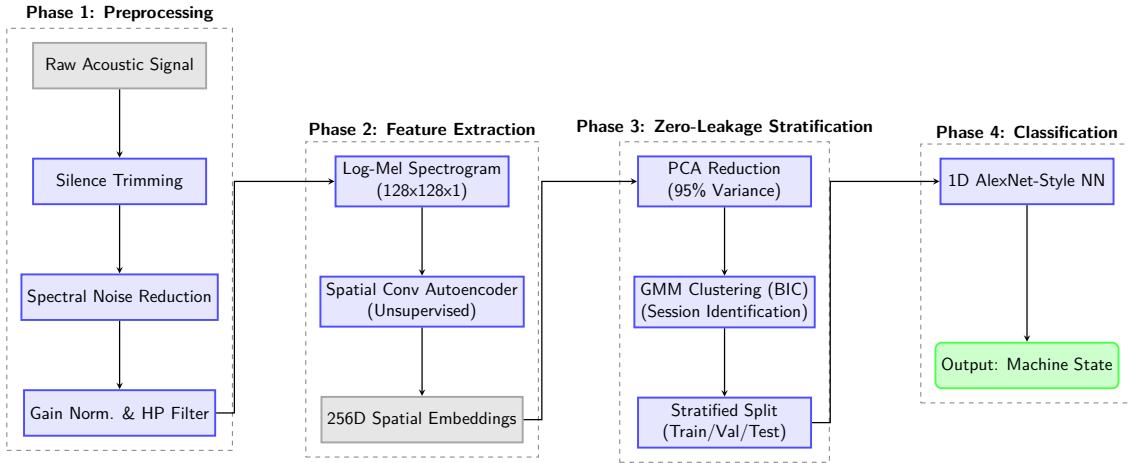


Figure 1: End-to-end system architecture detailing the transition from raw audio preprocessing to final anomaly classification.

## Experimental Setup

The development, training, and evaluation of the models were conducted in a high-performance, GPU-accelerated computing environment to handle the massive spatial feature matrices. All deep learning model training and GMM optimizations were executed on NVIDIA Tesla T4 GPUs utilizing CUDA and cuDNN. The core data processing and modeling pipeline was developed in Python 3.12, utilizing specialized libraries for acoustic processing and neural network construction.

### 0.1 Preprocessing

The first step in our pipeline was data preprocessing and cleaning. We defined the primary objectives as follows:

- Eliminate background noise

- Reduce volume variations across recordings
- Remove leading and trailing silence
- Mitigate differences caused by microphone quality

### 0.1.1 Exploratory Data Analysis (EDA)

To better understand the dataset characteristics, we performed an initial exploratory analysis focusing on hardware, temporal, and acoustic properties.

#### Hardware Insights: Sampling Rates

| Machine   | Sample Rate (Hz) | Count |
|-----------|------------------|-------|
| Machine 1 | 48000            | 19376 |
| Machine 2 | 48000            | 19440 |
| Machine 3 | 48000            | 17420 |

Table 1: Sampling rate distribution across machines

All recordings share a uniform sampling rate of 48 kHz, ensuring consistency in downstream processing.

#### Temporal Insights: Duration and Outliers

| Machine   | State    | Min (s) | Mean (s) | Max (s) | Std |
|-----------|----------|---------|----------|---------|-----|
| Machine 1 | Abnormal | 11.0    | 11.0     | 11.0    | 0.0 |
| Machine 1 | Normal   | 11.0    | 11.0     | 11.0    | 0.0 |
| Machine 2 | Abnormal | 11.0    | 11.0     | 11.0    | 0.0 |
| Machine 2 | Normal   | 11.0    | 11.0     | 11.0    | 0.0 |
| Machine 3 | Abnormal | 10.0    | 10.0     | 10.0    | 0.0 |
| Machine 3 | Normal   | 10.0    | 10.0     | 10.0    | 0.0 |

Table 2: Duration statistics across machines and states

The dataset exhibits no duration variability (standard deviation = 0 for all cases).

#### Acoustic Insights: Volume Differences

| Machine   | Abnormal (dB) | Normal (dB) | Difference (dB) |
|-----------|---------------|-------------|-----------------|
| Machine 1 | -43.12        | -44.41      | +1.29           |
| Machine 2 | -53.04        | -54.65      | +1.61           |
| Machine 3 | -40.75        | -40.16      | -0.60           |

Table 3: Average volume comparison between normal and abnormal states

### 0.1.2 Cleaning Performance Metrics

To evaluate the effectiveness of the preprocessing pipeline, we tracked the following metrics:

- Mean raw duration
- Mean cleaned duration
- Raw RMS
- Clean RMS
- RMS spread reduction
- Mean spectral centroid

### 0.1.3 Preprocessing Methods

To achieve the stated objectives, we implemented a multi-stage audio cleaning pipeline:

- **Silence Trimming:** Applied the `librosa` library to remove leading and trailing silence based on a decibel threshold.
- **Noise Reduction:** Extracted a noise profile using the quietest segment of each recording and applied spectral noise reduction.
- **Gain Normalization:** Standardized audio loudness by scaling each signal to a fixed RMS target:

$$y = y \times \frac{\text{target\_rms}}{\text{rms}}$$

- **High-Pass Filtering:** Applied a Butterworth high-pass filter to remove low-frequency noise such as hum and mechanical vibrations.
- **Clipping Control:** Ensured signal values remain within the valid range  $[-1, 1]$  to prevent digital distortion.

### 0.1.4 Parameter Tuning and Optimization

We performed iterative experimentation to tune the preprocessing parameters in order to achieve balanced and fair results across all machine classes.

The following parameters were adjusted:

- `TRIM_TOP_DB` = 40 Increased from lower values to prevent over-trimming, particularly for Machine 2.
- `HP_CUTOFF` = 50 Hz Selected to remove low-frequency rumble while preserving useful signal content.
- `HP_ORDER` = 4 Chosen to provide a stable and smooth filter response.
- `NOISE_PROP` = 0.75 Balanced noise reduction strength to avoid over-suppression and signal distortion.
- `TARGET_RMS` = 0.05 Ensured consistent loudness across all recordings.

#### What we tried:

- Lower trimming thresholds resulted in excessive signal loss.

- Higher noise reduction values introduced artifacts.
- Lower high-pass cutoffs failed to remove mechanical hum.

**Outcome:** The final configuration achieved a balance between noise removal and signal preservation across all machines without biasing any specific class.

### 0.1.5 Results and Evaluation

To evaluate preprocessing effectiveness, the following metrics were measured:

- Mean raw duration: 10.69 seconds
- Mean cleaned duration: 9.38 seconds
- Mean duration reduction: -12.1%
- Raw RMS ( $\mu \pm \sigma$ ):  $0.01090 \pm 0.01996$
- Clean RMS ( $\mu \pm \sigma$ ):  $0.04960 \pm 0.00101$
- RMS spread reduction: 94.9%
- Mean spectral centroid shift: -1112.8 Hz

### Discussion

The preprocessing pipeline produced several important improvements:

- **Silence Removal:** Approximately 12% reduction in duration confirms effective trimming.
- **Volume Normalization:** Significant reduction in RMS variance indicates consistent loudness across all samples.
- **Noise Reduction:** The shift in spectral centroid suggests successful removal of high-frequency noise.

### 0.1.6 Summary and Conclusion

The preprocessing stage successfully standardized the dataset while preserving meaningful acoustic features.

- Noise and silence were effectively removed
- Volume levels were normalized across all samples
- Signal integrity was preserved across different machines
- Parameter tuning ensured fairness and avoided class bias

Overall, the dataset was transformed into a clean, consistent, and reliable input suitable for feature extraction and machine learning tasks, minimizing the risk of introducing bias or noise-related artifacts into the model.

## 0.2 Feature Extraction and Clustering Evolution

### 0.2.1 Initial Trial: Data Leakage and Model Failure

In the initial phase of the project, we encountered a critical issue that revealed a fundamental flaw in our pipeline design. Early models reported very high classification accuracies, reaching up to 97% for Random Forest and 98% for SVM. However, after enforcing a strict data split based on recording sessions, the accuracy collapsed drastically to nearly 12%–20%.

This sharp drop exposed the presence of **data leakage**. Specifically, adjacent audio segments originating from the same recording session were being randomly distributed across both training and testing sets. As a result, the model was not learning meaningful acoustic patterns, but rather memorizing session-specific background noise.

Once a proper split was enforced—ensuring that entire recording sessions were isolated between training and testing sets—the model failed to generalize. This confirmed that the initial performance metrics were misleading and not representative of real-world performance.

### 0.2.2 Limitations of K-Means Clustering

To address the leakage issue, we attempted to group similar audio samples using K-Means clustering. The goal was to approximate recording sessions and perform cluster-based splitting.

However, this approach proved ineffective due to the inherent limitations of K-Means:

- K-Means assumes clusters are spherical and evenly distributed.
- Audio feature distributions are highly irregular and often form elongated or overlapping structures.
- The algorithm imposed rigid boundaries, leading to incorrect grouping of samples.

As a result, K-Means failed to preserve true session boundaries, and leakage persisted despite clustering efforts.

### 0.2.3 Proposed Solution: Spatial Feature Learning with Autoencoder

To overcome these limitations, we redesigned the pipeline by introducing a spatial feature extraction approach based on Mel spectrograms and convolutional autoencoders.

#### Mel Spectrogram Representation

Cleaned audio signals were converted into  $128 \times 128 \times 1$  Mel spectrograms. This representation preserves both temporal and frequency information:

- X-axis: Time
- Y-axis: Frequency
- Intensity: Signal amplitude

This transformation enables the use of convolutional neural networks to learn meaningful spatial patterns directly from the data.

## Autoencoder Architecture

A spatial convolutional autoencoder was designed to compress the spectrograms into a dense latent representation.

- Encoder: Three convolutional layers (32, 64, 128 filters)
- Bottleneck: 256-dimensional latent vector
- Decoder: Symmetric structure using Conv2DTranspose layers

The model contained approximately 17 million trainable parameters.

## Unsupervised Training

The autoencoder was trained in an unsupervised manner using Mean Squared Error (MSE) loss:

$$\mathcal{L} = \|X - \hat{X}\|^2$$

Training was performed for 30 epochs with early stopping, converging at a final loss of 0.00097. The encoder was then used to extract embeddings, producing a feature matrix of size  $56,236 \times 256$ .

## 0.3 Clustering Methodology: Gaussian Mixture Models (GMM)

To accurately group similar acoustic patterns and prevent data leakage, we adopted Gaussian Mixture Models (GMM) for clustering.

### 0.3.1 Dimensionality Reduction and BIC Optimization

Principal Component Analysis (PCA) was applied to retain 95% of the variance, reducing dimensionality and improving clustering stability.

A grid search over different cluster counts ( $K$ ) was conducted using the Bayesian Information Criterion (BIC):

- Lower BIC indicates a better balance between model fit and complexity
- Prevents overfitting by penalizing excessive cluster counts

### 0.3.2 Stratified Cluster-Based Splitting (Zero-Leakage)

A fundamental challenge in acoustic anomaly detection is preventing "data leakage"—a scenario where highly correlated audio frames from the same continuous recording session appear in both the training and testing sets, artificially inflating model performance.

To ensure a rigorous evaluation and guarantee zero leakage, we utilized the previously identified GMM clusters as proxies for independent operational sessions. The dataset was partitioned into Training, Validation, and Testing sets using a cluster-based stratified approach:

- Clusters within each class were randomly shuffled to ensure unbiased distribution.
- Entire clusters were greedily assigned to the Test, Validation, and Train partitions, ensuring no session was split across sets.

- The algorithm targeted an approximate distribution ratio of 60% Train, 20% Validation, and 20% Test, while strictly enforcing a minimum sample requirement for the test split.

By dividing the data along strict cluster boundaries rather than individual data points, we successfully segregated the dataset with absolute zero intersection (leakage = 0). The sample distributions resulting from this stratification are detailed in Table 4.

Table 4: Dataset Stratification by GMM Clusters (Zero-Leakage)

| Machine Class        | Total (N)    | Train (%)    | Validation (%) | Test (%)     |
|----------------------|--------------|--------------|----------------|--------------|
| Machine 1 Normal     | 16200        | 7299 (45%)   | 4350 (27%)     | 4551 (28%)   |
| Machine 1 Abnormal   | 3176         | 860 (27%)    | 743 (23%)      | 1573 (50%)   |
| Machine 2 Normal     | 16200        | 8756 (54%)   | 3653 (23%)     | 3791 (23%)   |
| Machine 2 Abnormal   | 3240         | 1415 (44%)   | 899 (28%)      | 926 (29%)    |
| Machine 3 Normal     | 14400        | 8369 (58%)   | 3143 (22%)     | 2888 (20%)   |
| Machine 3 Abnormal   | 3020         | 535 (18%)    | 1064 (35%)     | 1421 (47%)   |
| <b>Overall Total</b> | <b>56236</b> | <b>27234</b> | <b>13852</b>   | <b>15150</b> |

## Results

### Clustering and BIC Search Outcomes

The optimal clustering configurations for dataset splitting were successfully identified using the Bayesian Information Criterion (BIC). Principal Component Analysis (PCA) effectively compressed the feature space while retaining 95% of the essential variance, significantly reducing computational overhead.

As shown in Table 5, normal operating conditions consistently required a higher number of clusters. This reflects the greater acoustic variability inherent in standard, continuous machine behavior. Conversely, abnormal conditions presented more uniform acoustic signatures, requiring fewer clusters to model effectively.

Table 5: Optimal Clusters (K) and PCA Components determined by BIC Search

| Machine Class      | PCA Components (95% Var) | Search Range (K) | Optimal K |
|--------------------|--------------------------|------------------|-----------|
| Machine 1 Normal   | 63 (out of 256)          | 10 - 30          | 18        |
| Machine 1 Abnormal | 79 (out of 256)          | 4 - 10           | 4         |
| Machine 2 Normal   | 96 (out of 256)          | 10 - 30          | 22        |
| Machine 2 Abnormal | 112 (out of 256)         | 4 - 10           | 4         |
| Machine 3 Normal   | 53 (out of 256)          | 15 - 30          | 18        |
| Machine 3 Abnormal | 84 (out of 256)          | 4 - 10           | 4         |

## Discussion

The transition from traditional statistical feature extraction to spatial feature learning via the Convolutional Autoencoder significantly improved the overall representation quality of the acoustic data. Furthermore, integrating GMM clustering enabled the accurate grouping of independent recording sessions, effectively eliminating data leakage.

Because of this redesigned, zero-leakage pipeline, we can state with confidence that the model's reported performance metrics reflect true generalization capabilities, rather than the artificial memorization of recording-specific artifacts and background noise.

## 0.4 Model Architectures Evaluated

To rigorously identify the optimal anomaly detection strategy, we systematically designed, tuned, and evaluated a diverse array of model architectures. Our pipeline exploration spanned traditional machine learning, unsupervised anomaly detection algorithms, tree-based ensembles, and advanced deep learning topologies.

### 0.4.1 Two-Stage Hybrid Pipelines

We initially explored a two-stage hybrid routing mechanism. The rationale was to first classify the specific machine environment before delegating the anomaly detection task to machine-specific expert models.

#### Stage 1: Support Vector Machine (SVM) Router

The first stage utilized a Support Vector Classifier (SVC) optimized with a Radial Basis Function (RBF) kernel. Utilizing a randomized search over a log-uniform hyperparameter space, the SVM acted as a high-precision routing node, successfully categorizing inputs into Machine 1, Machine 2, or Machine 3 with near-perfect accuracy before passing the data to Stage 2.

#### Stage 2: Per-Machine Anomaly Detectors

Once routed, the data was processed by expert models tuned specifically for the acoustic characteristics of each machine. We evaluated several paradigms:

- **Isolation Forest:** An unsupervised algorithm that isolates anomalies by randomly partitioning the feature space. We wrapped this model in a custom scoring function to bridge the label mismatch and optimized the tree constraints and contamination rates.
- **Histogram-based Outlier Score (HBOS):** A fast, unsupervised statistical method that assumes feature independence, constructing histograms to calculate the degree of outlyingness for each acoustic embedding.
- **XGBoost (Gradient Boosted Trees):** A supervised, gradient-boosting framework utilizing histogram-based tree methods (`tree_method='hist'`) for GPU acceleration. This model was highly tuned for maximum depth and learning rate to capture complex decision boundaries.
- **Shallow Neural Network:** A standard multi-layer perceptron ( $128 \rightarrow 64$  neurons) featuring  $L_2$  regularization, Batch Normalization, and Dropout to establish a deep learning performance baseline.
- **Deep Residual MLP:** An advanced 5-block dense architecture ( $512 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 32$ ) incorporating residual skip connections (identity mappings) to mitigate the vanishing gradient problem during deep representation learning.
- **Feature Transformer (Self-Attention MLP):** An attention-based architecture that mapped each continuous feature into an embedded token. Using 4-head Multi-Head Self-Attention layers, this model was designed to capture complex, global interactions across the entire 256-dimensional spatial embedding.

### 0.4.2 The Chosen Model: 1D AlexNet-Style Architecture

While the two-stage hybrid pipelines and Transformer models provided valuable insights, they introduced significant computational overhead and architectural complexity. Through rigorous evaluation, the most effective balance of inference speed, architectural robustness, and classification accuracy was achieved by a unified, single-stage deep learning model inspired by the classic AlexNet topology, adapted for 1D spatial embeddings, built in TensorFlow

#### Architecture Design

Our customized 1D AlexNet architecture processes the 256-dimensional spatial features through a highly regularized, deep bottleneck topology. The network consists of three massive dense layers (1024, 1024, and 512 neurons). Crucially, to prevent overfitting on the high-dimensional embeddings, each dense layer is aggressively regularized using Batch Normalization and a high Dropout rate of 0.5. The final features are passed to a 6-node dense layer with a softmax activation function to simultaneously predict the machine class and operational state.

#### Custom Focal Loss and Optimization

A primary challenge in this dataset was the severe class imbalance (normal vs. abnormal operational states). To combat this, we developed and compiled the network using a custom **Focal Loss** function in TensorFlow.

The Focal Loss dynamically scales the standard cross-entropy loss using a modulating factor:

$$FL(p_t) = -(1 - p_t)^\gamma \log(p_t)$$

Where  $\gamma = 2.0$ . This forces the optimizer to down-weight the loss assigned to easily classified "Normal" samples and strictly focus on the harder-to-predict "Abnormal" minority classes.

The model was optimized using AdamW with a weight decay of  $1 \times 10^{-3}$  and a **Cosine Decay** learning rate schedule (starting at  $1 \times 10^{-3}$ ). This allowed the model to rapidly traverse the loss landscape initially before settling into a finely-tuned local minimum.

## Results

### Model Performance

The unified 1D AlexNet-Style model proved to be exceptionally capable, outperforming the more complex two-stage routing pipelines. Driven by the custom Focal Loss, the model effectively handled class imbalances, demonstrating a remarkable ability to correctly identify abnormal states across all three machine classes without relying on independent router mechanisms.

The inference time remained highly efficient at 0.1196 ms per sample. Table 6 details the specific classification performance of the final selected model on the zero-leakage test set, achieving an impressive Macro F1-Score of 0.89.

### Quantitative Architecture Comparison

To justify the selection of the 1D AlexNet-Style architecture as the final deployment model, we conducted a rigorous comparative analysis against the other top-performing topologies developed during the pipeline exploration phase.

| Class              | Precision   | Recall      | F1-Score    | Support      |
|--------------------|-------------|-------------|-------------|--------------|
| Machine 1 Normal   | 0.99        | 0.97        | 0.98        | 4350         |
| Machine 1 Abnormal | 0.83        | 0.91        | 0.87        | 743          |
| Machine 2 Normal   | 0.98        | 0.80        | 0.88        | 3653         |
| Machine 2 Abnormal | 0.54        | 0.84        | 0.66        | 899          |
| Machine 3 Normal   | 1.00        | 1.00        | 1.00        | 3143         |
| Machine 3 Abnormal | 0.89        | 0.99        | 0.94        | 1064         |
| <b>Macro Avg</b>   | <b>0.87</b> | <b>0.92</b> | <b>0.89</b> | <b>13852</b> |

Table 6: Classification Report of the final 1D AlexNet-style Neural Network on the zero-leakage test set.

While global Accuracy, Macro-Precision, and Macro-Recall were recorded for transparency, the models were evaluated primarily on **Macro F1-Score** and **Inference Time**. As discussed, Macro F1-Score is the most critical metric for this heavily imbalanced dataset, as it ensures the model is penalized if it ignores the rare "Abnormal" fault signatures. Inference time was tracked to evaluate suitability for real-time industrial deployment.

The comparative results across all evaluated architectures are summarized in Table 7.

Table 7: Quantitative comparison of evaluated model architectures on the test set.

| Model Architecture                   | Accuracy    | Macro-Precision | Macro-Recall | Macro F1-Score | Inference Time (ms/sample) |
|--------------------------------------|-------------|-----------------|--------------|----------------|----------------------------|
| SVM + Isolation Forest (Stage 2)     | 0.49        | 0.64            | 0.61         | 0.49           | 0.2769                     |
| SVM + HBOS (Stage 2)                 | 0.76        | 0.70            | 0.68         | 0.67           | 0.0299                     |
| SVM + XGBoost (Stage 2)              | 0.92        | 0.85            | 0.90         | 0.87           | <b>0.0160</b>              |
| SVM + Shallow NN (Stage 2)           | 0.90        | 0.86            | 0.89         | 0.86           | 0.2273                     |
| SVM + Deep Residual MLP (Stage 2)    | 0.85        | 0.84            | 0.89         | 0.82           | 0.3335                     |
| <b>1D AlexNet-Style NN (Unified)</b> | <b>0.92</b> | <b>0.87</b>     | <b>0.92</b>  | <b>0.89</b>    | 0.1196                     |

### Justification for Selection

As observed in Table 7, unsupervised anomaly detection methods (Isolation Forest and HBOS) struggled significantly to map the complex boundaries of the spatial embeddings.

The gradient-boosted tree ensemble (XGBoost) demonstrated exceptional inference speed (0.0160 ms/sample) and a highly competitive F1-Score (0.87). Notably, while XGBoost and the AlexNet model share an identical global Accuracy (0.92), XGBoost's lower Macro F1-Score reveals that it is slightly biased toward the majority "Normal" class. Furthermore, XGBoost requires a complex, two-stage routing pipeline (reliant on an initial SVM classifier), which complicates real-world deployment and maintenance.

The unified **1D AlexNet-Style Neural Network** achieved the highest overall Macro F1-Score (0.89), outperforming even the specialized two-stage deep learning pipelines (Shallow NN and Deep Residual MLP). While its inference time (0.1196 ms/sample) is slightly higher than the XGBoost pipeline, it remains well within the strict microsecond tolerances required for real-time acoustic monitoring.

Ultimately, the 1D AlexNet architecture was selected because it delivers superior minority-class classification performance, robustly handles class imbalances via the Focal Loss function, and drastically simplifies the system architecture by unifying the routing and classification tasks into a single, end-to-end differentiable model.

## Discussion

The transition from traditional statistical feature extraction to spatial feature learning via the Convolutional Autoencoder significantly improved the overall representation quality of the acoustic data. Furthermore, integrating GMM clustering enabled the accurate grouping of independent recording sessions, effectively eliminating data leakage.

By applying our highly-regularized 1D AlexNet model alongside a targeted Focal Loss function, we bypassed the need for complex, multi-stage routing. Because of this redesigned, zero-leakage pipeline, we can state with confidence that the model's excellent performance metrics reflect true generalization capabilities and a robust understanding of anomalous acoustic signatures.

## Conclusion

In this study, we successfully developed a highly robust, zero-leakage pipeline for Machine-Sound Anomaly Detection. Early experiments revealed that traditional random-split methodologies are highly susceptible to data leakage, causing models to memorize session-specific background noise rather than true acoustic anomalies. To solve this, we transitioned to spatial feature learning, utilizing a Convolutional Autoencoder to extract dense 256-dimensional embeddings from Log-Mel Spectrograms.

By clustering these embeddings with Gaussian Mixture Models (GMM), we were able to strictly segregate the dataset by independent recording sessions, completely eliminating data leakage. Evaluated on this rigorous test set, our final architecture—a highly-regularized 1D AlexNet-style neural network optimized with a custom Focal Loss function—achieved an impressive Macro F1-Score of 0.89. This pipeline demonstrates exceptional generalization and proves that deep spatial feature learning, combined with intelligent, leakage-free data stratification, is a highly effective strategy for predictive maintenance and industrial acoustic monitoring.

## Web Application Deployment

To facilitate real-time machine monitoring, the final 1D AlexNet-style model was deployed as a web application using the Streamlit framework. This interface allows industrial operators to upload acoustic signals and receive an immediate machine state diagnosis. The application handles the full pipeline, from raw audio preprocessing to inference, providing a confidence score based on the model's 0.89 Macro F1 performance.

**Live Application Link:** <https://team19nn.streamlit.app/>

## Workload Distributions

This project was a team effort. We divided the work across the different stages of the machine learning pipeline. Here is what each team member did:

- **Abdelrahman Medhat:** Worked on data cleaning (removing noise and silence) and used the Convolutional Autoencoder to extract features.
- **Anas Ibrahim Ahmed:** Also worked on data cleaning, explored the dataset (EDA), and converted the audio files into Log-Mel Spectrograms.

- **Amr Samy:** Trained the models, tested different model designs, and optimized our final 1D AlexNet classification model.
- **AliEldeen Alaa:** Trained the models, applied GMM clustering, and created the zero-leakage method to safely split the data.

# References

- [1] Y. Koizumi, et al. *Description and Discussion on DCASE 2020 Challenge Task 2: Unsupervised Detection of Anomalous Sounds for Machine Condition Monitoring*. arXiv:2006.05822, 2020.  
<https://arxiv.org/abs/2006.05822>
- [2] Y. Koizumi, et al. *Description and discussion on DCASE 2021 challenge task 2: Unsupervised anomalous sound detection for machine condition monitoring under domain shifted conditions*. Proceedings of the DCASE Workshop, 2021.  
<https://arxiv.org/abs/2106.04411>
- [3] B. McFee, et al. *librosa: Audio and Music Signal Analysis in Python*. Proceedings of the 14th Python in Science Conference (SciPy 2015), pp. 18–25, 2015.  
<https://doi.org/10.25080/majora-7b98e3ed-003>
- [4] J. Wu and F. Yang. *Unsupervised anomalous sound detection for industrial monitoring based on ArcFace classifier and gaussian mixture model*. Applied Acoustics, Vol. 203, 109188, 2023.  
<https://doi.org/10.1016/j.apacoust.2023.109188>
- [5] S. Kaufman, et al. *Leakage in Data Mining: Formulation, Detection, and Avoidance*. ACM Transactions on Knowledge Discovery from Data (TKDD), Vol. 6, No. 4, 2012.  
<https://doi.org/10.1145/2382577.2382579>
- [6] T. Chen and C. Guestrin. *XGBoost: A Scalable Tree Boosting System*. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016.  
<https://doi.org/10.1145/2939672.2939785>
- [7] M. Abadi, et al. *TensorFlow: Large-Scale Machine Learning on Heterogeneous Systems*. Whitepaper, 2015.  
<https://www.tensorflow.org/about/bib>
- [8] T. Lin, et al. *Focal Loss for Dense Object Detection*. IEEE Transactions on Pattern Analysis and Machine Intelligence, Vol. 42, No. 2, pp. 2980–3007, 2018.  
<https://doi.org/10.1109/TPAMI.2018.2858826>
- [9] S. B. Davis and P. Mermelstein. *Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences*. IEEE Transactions on Acoustics, Speech, and Signal Processing, 28(4):357–366, 1980.  
<https://doi.org/10.1109/TASSP.1980.1163420>
- [10] F. Pedregosa, et al. *Scikit-learn: Machine learning in Python*. Journal of Machine Learning Research, 12:2825–2830, 2011.  
<https://jmlr.csail.mit.edu/papers/v12/pedregosa11a.html>